Evaluations Are Crossed Designs: A Hierarchical Bayesian Interaction Model for Analyzing Model Evaluations

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Abstract. Most language-model evaluations report a mean score, sometimes with a standard error. This discards three things the data contains: the scores are usually graded rather than binary, the design is models crossed with items, and models have item-specific strengths, a model-by-item interaction. We argue that the appropriate analysis is a hierarchical Bayesian graded-response model with crossed random effects for models and items and an explicit model-by-item interaction term, fit by MCMC. This sits in the explanatory item-response and generalizability-theory traditions but fills a gap: existing item-response work in machine learning is almost entirely binary, variational, and aimed at subselecting items rather than analyzing the systems under test. We demonstrate the model on a multi-turn instruction-retention evaluation, where a ladder of models shows the interaction term improving out-of-sample fit by about 59 ELPD (five standard errors) over additive specifications, while additive per-model terms add nothing. Reporting raw means or additive effects on this data misattributes interaction variance to main effects and manufactures a spurious finding that the interaction model dissolves. We give a concrete specification, priors, and a reporting checklist, and discuss when the model is and is not worth its cost.

Keywords: model evaluation , item response theory , Bayesian inference , generalizability theory , measurement

1 Introduction

A language-model evaluation produces a table: systems in rows, test items in columns, a score in each cell. The near-universal summary is a column mean per system, occasionally with a standard error (Miller, 2024). This summary throws away most of the structure in the table. First, the scores are frequently *graded*: a rubric returns 0–3, a judge returns a Likert rating, a partial-credit checker returns degrees of compliance. Averaging treats these as interval-scaled, which they are not. Second, the design is *crossed*: every system meets every item, so item difficulty and system ability are confounded in any per-system mean, and a system that draws an easier item subset looks better for free. Third, and most consequentially, systems have *item-specific strengths*: a model strong on one task may be weak on another in a way not predicted by its average. This is a model-by-item interaction, and a mean by construction cannot see it.

We argue for a specific analysis as a default: a hierarchical Bayesian graded-response model with crossed random effects for systems and items and an explicit model-by-item interaction term, estimated by MCMC. The ingredients are standard in psychometrics, item response theory (IRT) (De Boeck & Wilson, 2004; Samejima, 1969), crossed-effects modeling (Baayen et al., 2008), and generalizability theory (Brennan, 2001), and are well supported by modern tooling (Abril-Pla et al., 2023; Bürkner, 2021; McElreath, 2020). What is new is assembling them for the evaluation setting: treating evaluation scores as ordinal, putting the experimental manipulation as a covariate on ability, and making the model-by-item interaction a first-class, identifiable term.

Our contributions are: (i) we characterize evaluation data as a crossed, graded, interaction-bearing design and state the model that matches it; (ii) we show, on a real evaluation, that the interaction term is the dominant structure, $\approx$59 ELPD better than additive models, and that ignoring it (raw means or additive effects) produces a concrete, published error; (iii) we give a reproducible specification, including why an interaction variance component is preferable to a low-rank factor model on identifiability grounds; and (iv) we provide a reporting checklist and an honest account of the method's costs.

2 What the field does, and the gap

Three lines of work move beyond the raw mean, each addressing one facet but not the whole.

Uncertainty. Miller (2024) treats an eval as an experiment and imports standard errors, clustering, and paired differences; Bowyer et al. (2025) shows the normal approximation underestimates uncertainty at the small sample sizes typical of specialized evals. This fixes the error bar on a mean but keeps the mean: no item structure, no interaction.

Item response theory. IRT has been used to build evaluation scales (Lalor et al., 2016), reweight leaderboards by item informativeness (Rodriguez et al., 2021), compare test sets (Vania et al., 2021), and construct efficient benchmarks (Polo et al., 2024). This literature is almost entirely *binary* (correct/incorrect), estimated by *variational* approximation for scale, and aimed at *item subselection* rather than analyzing the systems under test. The one graded-response application we found diagnoses judge reliability, not the systems being evaluated (Choi et al., 2026).

Measurement and ranking. Construct-validity work asks whether benchmarks measure what they claim (Bean et al., 2025); generalizability theory decomposes score variance across facets but in an ANOVA, non-generative form (Song et al., 2025); preference ranking uses Bradley–Terry with intervals (Chiang et al., 2024) but for pairwise outcomes, not graded item scores. LLM-as-judge reliability is typically reported as raw agreement, which is not chance-corrected (Zheng et al., 2023).

The gap is the intersection: no standard approach simultaneously (a) treats scores as *ordinal*, (b) models systems and items as *crossed with partial pooling*, and (c) includes an explicit *model-by-item interaction*. The pieces exist in adjacent literatures; assembling them is the contribution.

3 The recommended model

Let $y_i \in \{0, \dots, K\}$ be the graded score for observation i with system $m[i]$, item $d[i]$, and an optional experimental covariate t_i (a condition, intervention, or treatment). The model is a cumulative-link (ordered-logistic) graded-response model,

$$\Pr(y_i \le k) = \text{logistic}(\kappa_k - \eta_i), \qquad \kappa_1 < \cdots < \kappa_K,$$

with linear predictor

$$\eta_i = \alpha_{d[i]} + \beta_{d[i]}\, t_i + g_{m[i]} + s_{m[i]}\, t_i + z_{m[i],d[i]}\, t_i.$$

Here α_d is per-item baseline difficulty, β_d the per-item effect of the covariate, g_m the per-system ability, s_m the per-system covariate sensitivity, and $z_{m,d} \sim \mathcal{N}(0, \sigma_{\text{int}})$ the **model-by-item interaction**. All group effects are non-centered; cutpoints use an offset parameterization; SDs take weakly-informative half-normal/exponential priors and correlations, where used, an LKJ prior (Bürkner, 2021; McElreath, 2020).

Four choices deserve justification. *Ordinal likelihood*: graded scores are not interval-scaled, so a cumulative link is the principled choice over a Gaussian or a binarization. *Crossed effects*: systems and items are fully crossed, so crossed random intercepts remove the confound between item difficulty and system ability that a per-system mean carries; this also yields a generalizability-theory variance decomposition for free (Brennan, 2001). *Interaction as a variance component*: the non-additivity can be written as a low-rank product $\lambda_m \varphi_d$ (a factor / 2PL form), but that is unidentified up to sign and scale and in practice multimodal; the additive cell effect $z_{m,d} \sim \mathcal{N}(0, \sigma_{\text{int}})$ is fully identified, samples cleanly, and gives a single interpretable estimand, the interaction magnitude. *Full Bayesian estimation*: MCMC propagates uncertainty into every downstream quantity, including system rankings, where overlapping intervals correctly read as "tied"; partial pooling shrinks noisy per-cell estimates, the behavior that makes small-n evals tractable.

What the model returns that a mean does not: per-item and per-system effects with calibrated intervals; a variance decomposition (how much of the variation is item, system, interaction); model comparison by leave-one-out cross-validation (Vehtari et al., 2017) to test whether structure is warranted; and, for judge-scored evals, a place to put rater effects and to report chance-corrected agreement rather than raw agreement.

Table 1. Leave-one-out comparison on the retention evaluation. Models with a model-by-item interaction (M8, M_irt) fit far better than additive models.

Model	Structure	ELPD	ΔELPD (SE)
M8	+ model×item interaction (variance)	−545.9	—
M_irt	+ 2PL discrimination $a_d\theta_m$	−555.4	9.5 (10.5)
M_corr	additive + correlated slopes	−604.4	58.5 (11.2)
M4	additive per-model + per-item	−604.8	58.9 (11.2)
M0	per-item only (no system structure)	−607.2	61.3 (11.7)

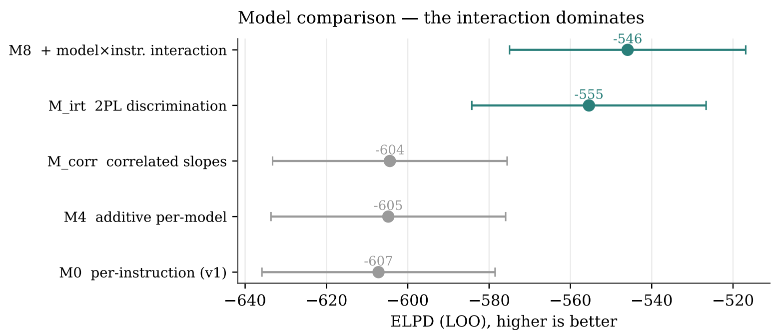

Fig. 1. Out-of-sample fit (ELPD) with standard errors. The two models that include a model-by-item interaction (teal) separate cleanly from the additive models (grey), which are indistinguishable from each other.

4 Case study: the interaction is the structure

We apply the framework to a multi-turn instruction-retention evaluation (Korenblit, 2026): five models are scored on whether they retain twelve coding instructions across three codebases, under a baseline and a treatment (stating the instruction), on a 0–3 compliance scale, 1,067 observations. We fit a ladder of increasingly structured models and compare them by leave-one-out cross-validation (Table 1, Fig. 1).

The result is unambiguous. The two specifications that capture a model-by-item interaction fit decisively better than every additive model, by about 59 ELPD, more than five standard errors. The two interaction forms (an interaction variance component, and an IRT discrimination term) are indistinguishable from each other, so the interaction matters, not its exact parameterization. The additive embellishments do nothing: a per-system main effect (M4) is indistinguishable from a model with no system structure at all (M0), and correlating system intercepts with slopes (M_corr) does not help. Under M8 the interaction is the largest structured source of variance, $\sigma_{\text{int}} = 1.84$ (94% HDI $[1.26, 2.40]$), exceeding the per-item, per-system, and per-system-slope components.

This is not a curiosity; it changes published conclusions. The mean and the additive model attribute the interaction variance to main effects. In this dataset that inflated the per-item effect (its SD nearly halves when the interaction is admitted) and, combined with a scoring bug, produced a reported finding that one instruction was *harmed* by being stated. The interaction model localizes that effect to a single extreme (system, item) cell of one model, and a separate framing experiment confirms it is an artifact, not a real effect. An analysis that cannot represent the interaction cannot diagnose this.

5 A reporting checklist

For an evaluation of systems crossed with graded items we recommend: (1) model scores with an ordinal/cumulative link, not a mean over a graded scale; (2) include crossed random effects for systems and items with partial pooling; (3) include a model-by-item interaction term, and report σ_{int} alongside the main-effect SDs; (4) compare specifications by leave-one-out cross-validation rather than asserting a structure; (5) report rankings with their posterior intervals, treating overlap as "tied"; (6) check prior sensitivity of the variance components, which are prior-influenced when the number of systems or items is small; (7) for judge-scored evals, model rater effects and report chance-corrected agreement. Items (1)–(3) are the substance; (4)–(7) are the discipline that keeps the conclusions honest.

6 Limitations and when not to use this

The variance components are prior-sensitive when groups are few; with a handful of systems or items, report rankings and the qualitative interaction conclusion, not precise variance magnitudes. MCMC is more expensive than the variational estimation common in IRT-for-eval work; for very large item banks aimed at subselection, amortized variational methods remain appropriate, and the model here is better suited to analyzing a fixed set of systems than to scaling to millions of items. When scores are genuinely binary and items are not crossed with systems in a way that admits interaction, the full model reduces to simpler, well-understood special cases, and the added structure is not worth its cost. The recommendation is for the common middle case: a moderate number of systems, graded items, a crossed design, and a comparative question.

7 Conclusion

Evaluation tables are crossed designs of graded items, and the quantity that matters is often how systems differ item by item, an interaction a mean cannot represent. A hierarchical Bayesian graded-response model with crossed effects and a model-by-item interaction matches this structure, is supported by established psychometric theory and modern tooling, and, on a real evaluation, reveals structure that additive analyses miss by 59 ELPD and that raw means turn into a spurious finding. We recommend it as a default for evaluation analysis and provide the specification and checklist to apply it.

Use of large language models. During the preparation of this work the author used a large language model (Claude) to improve the language and readability of the manuscript and to assist with analysis and figure code; all results were computed by the statistical models and verified by the author.

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